

Gravity model of interregional migration

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1 Motivations

- Le projet Navigate - Modèle initial
- Littérature

2 Base de données

- Construction de la base de données
- Faits stylisés

3 Analyse économétrique

- Méthodologie
- Résultats
- Tests de robustesse

4 Conclusion

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4 Conclusion

Motivations : MEI avec démographie

Les Modèles d'Evaluation Intégrée (IAM) :

- Modèles numériques intégrant des modèles simplifiés de l'économie, du système de production ainsi que des sources d'énergie en vue de prédire les émissions carbone et donc l'évolution du climat
- Dynamiques démographiques ignorées dans les premières versions évolutions exogènes

Voir : [Nordhaus, 2008]

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Voir : [Nordhaus, 2008]

Projet Navigate :

- Améliorer les modèles IAM
- Groupe de travail 4 : hétérogénéité spatiale et sociale des effets du changement climatique
- Post-doctorat : intégration des dynamiques démographiques, migration

Motivations : le projet

Point de départ : *Climate change and population : An assessment of mortality due to health impacts*, Pottier A., Fleurbaey M., Méjean A., Zuber S.

- Objectifs : prise en compte de la surmortalité en fonction des classes d'âge
- Méthodes : méthode des cohortes, modèle IAM
- Données : estimations de la population par le World Population Prospects (UN), estimations de la surmortalité par l'OMS [WHO, 2014]

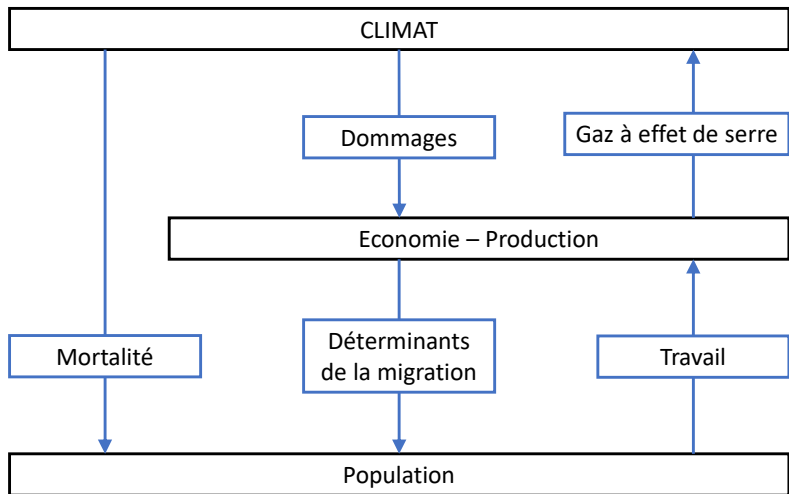
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Objectif de ma contribution : Intégrer la migration dans le modèle

- Contraintes
 - Prédictions des flux migratoires adaptées à la méthode des cohortes
 - Données endogènes disponibles : PIB, population



Littérature sur les modèles gravitaires I

Fondation théorique : RUM Model : Random Utility Maximization model : voir [Beine et al., 2016]

- Utilité tirée par un individu dans un pays en comparaison de celle reçue ailleurs
- Bénéfices (gain en termes de revenu) et coûts (distances)
- Facteurs non observés
- Résistance multilatérale

Données sur la migration : voir [Abel et al., 2019] or [Abel and Cohen, 2019]

- Les différents types de données
 - Mesures dans le pays de destination
 - Mesures dans le pays envoyeur
 - Comptabilité démographique
- Migration illégale
- Les autres types de migrations

Littérature sur les modèles gravitaires II

Résultats clés : voir [Beine et al., 2016]

- Facteurs spécifiques à l'origine ou à la destination (non-linéaire ou non-monotone) :
 - Revenu (*par ex.* PIB par habitant, salaires, revenus spécifiques à l'éducation) : mesure l'attractivité de chaque lieu.
 - Contraintes de crédit (*e.g.* PIB par habitant dans le pays d'origine, pauvreté) : mesurent l'accessibilité
 - Attentes : les migrants fondent leur décision sur le revenu attendu dans le pays de destination plutôt que sur le revenu actuel.
 - Politiques générales d'immigration : ces politiques ont en effet un impact sur le coût potentiel de la migration.
- Facteurs dyadiques :
 - Réseaux (diaspora ou stock de migrants dans le pays d'accueil, liens coloniaux) : ↓ coût de la migration.
 - Politiques d'immigration bilatérales : ↑ coût de la migration.
 - Proximité linguistique et culturelle : ↓ coût de la migration.

Littérature sur le lien migration/changement climatique

Analyse de la migration et de l'environnement

- Analyse avec variables environnementales : température, précipitations, variabilité
- Analyse avec événements catastrophiques : cyclones, inondations, sécheresses
- Analyse via l'urbanisation : changement d'utilisation des sols

Résultats clés

- Résultats ambigus sur l'effet des variables environnementales
- Effets plus importants pour la migration entre zones rurales et urbaines

Voir : [Lilleor and den Broeck, 2011] et [Millock, 2015]

On adopte une approche très différente : fonction de dommage

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Sources des données

- Données dyadiques : CEPII, voir [Mayer et al., 2014]
(environ 230 pays)
- Données économiques : Banque mondiale, World Development Indicators
(environ 217 pays, annuelles de 1990-2019)
- Données migratoires : UNSD
(environ 230 pays, quinquennales 1990-1995-2000-2005-2010-2015-2020)
- Données sur les réfugiés : UNHCR
(environ 230 pays, annuelles, 1990-2020)
- Découpage en 20 régions selon la méthodologie du WPP

Structure du code I

Caractéristiques individuelles et dyadiques des pays

http://www.cepii.fr/CEPII/fr/bdd_modele/presentation.asp?id=8

Figure 1 – CEPII dataset

iso3_o	iso3_d	contig	comlang_off	comlang_ethno	colony	comcol	curcol	col45	smctry	distw	distwces	dist	distcap
ABW	ABW	0	0	0	0	0	0	0	0	25.09354	23.04723	5.225315	5.225315
ABW	AFG	0	0	0	0	0	0	0	0	13168.22000	13166.37000	13257.810000	13257.810000
ABW	AGO	0	0	0	0	0	0	0	0	9587.31600	9584.19300	9516.913000	9516.913000
ABW	AIA	0	0	0	1	0	0	0	0	976.89740	976.89160	983.268200	983.268200
ABW	ALB	0	0	0	0	0	0	0	0	9091.57600	9091.46600	9091.742000	9091.742000
ABW	AND	0	1	0	0	0	0	0	0	7570.08400	7570.08300	7572.788000	7572.788000
ABW	ANT	0	1	1	0	1	0	0	1	239.90640	142.85830	136.384800	136.384800
ABW	ARE	0	0	0	0	0	0	0	0	12773.08000	12772.95000	12735.010000	12735.010000
ABW	ARG	0	1	0	0	0	0	0	0	5187.78800	5157.12600	5396.220000	5396.220000
ABW	ARM	0	0	0	0	0	0	0	0	11106.96000	11106.76000	11107.780000	11107.780000
ABW	ATG	0	0	1	0	0	0	0	0	1011.00300	1010.36300	1024.995000	1024.995000
ABW	AUS	0	0	1	0	0	0	0	0	15670.46000	15638.10000	15237.500000	15354.320000
ABW	AUT	0	0	0	0	0	0	0	0	8632.97100	8631.02200	8719.130000	8719.130000
ABW	AZE	0	0	0	0	0	0	0	0	11421.98000	11420.91000	11503.000000	11503.000000
ABW	BDI	0	0	0	0	0	0	0	0	11138.74000	11138.61000	11120.470000	11120.470000
ABW	BEL	0	1	1	0	0	0	0	0	7843.25500	7843.00600	7847.070000	7847.070000
ABW	BEN	0	0	0	0	0	0	0	0	7915.63800	7915.21400	7979.945000	8006.697000
ABW	BFA	0	0	0	0	0	0	0	0	7348.56800	7346.22800	7414.016000	7414.016000

- Distances : distw - distcap
- Colony - Comlang_off - Contig
- Land - population - PIB en parité de pouvoir d'achat (source [WDI - Banque Mondiale](#))

⇒ Dataset countries_char

Structure du code II

Données bilatérales sur les stocks de migrants :

<https://www.un.org/development/desa/pd/content/international-migrant-stock>

Figure 2 – United Nations Statistics Division - Department of Economic and Social Affairs



United Nations

Department of
Economic and
Social Affairs

International Migrant Stock 2020: Destination and origin

Table 1: International migrant stock at mid-year by sex and by region, country or area of destination and origin, 1990-2020

POPSTAT2020-Rev.2020

Suggested citation: United Nations Department of Economic and Social Affairs, Population Division (2020). International Migrant Stock 2020

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Index	Region, development group, country or area of destination	Notes of destination	Location code of	Type of data of	Region, development group, country or area of origin	Location code of	International migrant stock at mid-year, both sexes combined						
							1990	1995	2000	2005	2010	2015	2020
1	WORLD		900		WORLD	900	152 986 157	161 389 976	173 230 585	181 448 028	220 983 187	247 963 644	280 586 105
2	WORLD		900		Sub-Saharan Africa	947	15 334 807	16 488 973	15 630 014	17 274 905	19 842 724	24 197 211	28 294 538
3	WORLD		900		Northern Africa and Western Asia	1833	16 365 218	18 869 437	19 437 054	21 867 791	25 313 711	32 608 047	37 563 820
4	WORLD		900		Central and Southern Asia	921	31 104 020	29 969 983	30 087 694	33 347 690	41 002 380	46 401 130	51 229 549
5	WORLD		900		Eastern and South-Eastern Asia	1832	14 774 419	17 889 923	21 440 854	25 556 983	30 877 286	35 205 590	38 488 740
6	WORLD		900		Latin America and the Caribbean	1830	13 272 399	19 469 754	24 030 780	29 333 298	34 637 616	39 209 000	42 286 481
7	WORLD		900		Oceania (excluding Australia and New Zealand)	1835	272 517	313 961	375 176	407 138	441 993	495 220	565 281
8	WORLD		900		Australia and New Zealand	927	689 030	758 234	871 660	984 280	1 149 265	1 256 202	1 484 924
9	WORLD		900		Europe and Northern America	1829	58 532 413	51 582 164	52 677 703	54 949 090	58 601 184	68 592 402	67 691 621
10	WORLD		900		Developed regions	901	51 826 248	52 993 691	54 458 296	56 551 036	60 587 161	62 613 012	69 815 370
11	WORLD		900		Less developed regions	902	92 519 577	109 816 788	111 760 589	127 074 327	151 630 795	174 420 790	186 125 584
12	WORLD		900		Least developed regions, excluding least developed countries	934	62 698 205	72 741 458	83 546 070	97 124 483	116 141 851	132 944 110	150 528 987
13	WORLD		900		Less developed regions, excluding China	948	67 652 367	95 081 133	105 043 210	118 876 440	141 938 473	163 093 380	186 511 434
14	WORLD		900		Least developed countries	941	29 823 372	29 117 338	28 062 489	29 949 099	35 497 699	41 482 680	47 684 597
15	WORLD		900		Land-locked Developing Countries (LLDC)	1836	25 088 015	23 874 777	23 134 479	24 288 062	27 740 965	31 336 289	35 277 768
16	WORLD		900		Small island developing States (SIDS)	1837	5 452 503	6 572 361	7 783 822	8 591 316	9 401 155	10 467 162	11 559 498
17	WORLD		900		High-income countries	1803	34 008 977	35 677 967	38 024 485	40 882 214	45 849 196	47 356 719	52 888 019
18	WORLD		900		Middle-income countries	1817	88 958 699	86 829 745	109 819 964	123 912 679	144 581 899	159 981 964	177 305 634
19	WORLD		900		Upper-middle-income countries	1802	46 251 718	53 859 271	69 585 818	85 735 241	74 175 415	77 576 140	87 848 518
20	WORLD		900		Lower-middle-income countries	1501	42 707 281	45 870 487	59 294 136	58 177 438	70 406 484	80 405 824	89 747 316
21	WORLD		900		Low-income countries	1500	21 142 440	18 959 403	18 002 695	18 095 623	21 710 616	31 399 947	37 419 672
22	WORLD		900		AFRICA	903	20 621 464	22 341 394	22 080 063	25 241 406	29 193 852	35 111 959	40 567 163
23	WORLD		900		Eastern Africa	949	7 704 743	7 774 774	6 679 982	6 192 747	7 524 192	8 024 290	8 527 289
24	WORLD		900		Middle Africa	911	1 822 243	1 187 032	2 439 380	2 728 656	3 007 542	3 916 990	4 426 662
25	WORLD		900		Northern Africa	912	5 296 657	5 852 421	6 441 989	7 966 421	9 251 128	10 914 740	12 282 628
26	WORLD		900		Southern Africa	913	576 752	664 456	732 312	904 014	1 199 813	1 129 951	1 278 435

Données bilatérales sur les réfugiés : source,
<https://www.unhcr.org/refugee-statistics>

Figure 3 – United Nations Refugee Agency


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[Methodology](#)

Dataset
Population figures

Display type
Totals

Population types
All

Year
2020, 2015, 2010, 2005, 2000, 1995, 1990

Country of Origin
All countries

Country of Asylum

☐ Do not display
☒ All countries
☐ Display by Region
☐ Custom selection

* The data for the latest year (2020) is available up until the mid-year. IDMC data on Internally Displaced Persons and the demographic data for the population and solutions datasets are not collected at mid-year and therefore only available up until year-end in the previous year (2019).

Query permalink

www.unhcr.org/refugee-statistics/download?url=neb8GY

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Year	Country of Origin	Country of Asylum	Refugees under UNHCR's mandate	Asylum-seekers	IDPs of concern to UNHCR	Venezuelans displaced abroad	Stateless persons	Others of concern
1990	Unknown (null)	Afghanistan (AFG)	50	0	0		0	0
1990	Palestinian (PSE)	Algeria (DZA)	4 000	0	0		0	0
1990	Western Sahara (ESH)	Algeria (DZA)	165 000	0	0		0	0
1990	Unknown (null)	Algeria (DZA)	110	0	0		0	0
1990	Dem. Rep. of the Congo (COD)	Angola (AGO)	9 212	0	0		0	0
1990	Namibia (NAM)	Angola (AGO)	245	0	0		0	0
1990	South Africa (ZAF)	Angola (AGO)	2 100	0	0		0	0
1990	Ethiopia (ETH)	Egypt (EGY)	605	0	0		0	0

Structure du code IV

Figure 4 – United Nations - Department of Economic and Social Affairs

Year	m49code_d	m49code_o	mig_stock
2020	900	900	280596105
2020	900	947	28284538
2020	900	1833	37563620
2020	900	921	51229549
2020	900	1832	38400740
2020	900	1830	42890481
2020	900	1835	565281
2020	900	927	1404924
2020	900	1829	67601621
2020	900	901	69815370
2020	900	902	198125584
2020	900	934	150520987
2020	900	948	186511434
2020	900	941	47604597
2020	900	1636	35277768
2020	900	1637	11559408
2020	900	1503	52808019
2020	900	1517	177395834
2020	900	1502	87648518

Figure 5 – United Nations - Department of Economic and Social Affairs

Year	countryname_o	iso3_o	countryname_d	iso3_d	refugees_UNHCR
1990	Afghanistan	AFG	China	CHN	50
1990	Afghanistan	AFG	Denmark	DNK	192
1990	Afghanistan	AFG	United Kingdom of Great Britain and Northern Ireland	GBR	237
1990	Afghanistan	AFG	India	IND	11946
1990	Afghanistan	AFG	Iran (Islamic Rep. of)	IRN	306110
1990	Afghanistan	AFG	Italy	ITA	394
1990	Afghanistan	AFG	Lebanon	LBN	300
1990	Afghanistan	AFG	Pakistan	PAK	3253000
1990	Afghanistan	AFG	Sweden	SWE	378
1990	Afghanistan	AFG	United States of America	USA	11488
1995	Afghanistan	AFG	Afghanistan	AFG	0
1995	Afghanistan	AFG	Egypt	EGY	23
1995	Afghanistan	AFG	Bulgaria	BGR	475
1995	Afghanistan	AFG	Canada	CAN	4575
1995	Afghanistan	AFG	Denmark	DNK	748
1995	Afghanistan	AFG	Ecuador	ECU	8
1995	Afghanistan	AFG	France	FRA	1294

Structure du code V

```

bilateral_list <- select(countries_char, c("Year", "m49code_o", "m49code_d")) %>% mutate (zeros=0)
bilateral_list$Year <- ifelse(bilateral_list$Year==2019, 2020, bilateral_list$Year)

bil_mig_UNSD <- merge(x=bilateral_list, y=bil_mig_UNSD, by =c("Year", "m49code_o", "m49code_d"), all=TRUE)
bil_mig_UNSD <- bil_mig_UNSD[!is.na(bil_mig_UNSD$zeros), ]
bil_mig_UNSD$mig_stock <- ifelse(is.na(bil_mig_UNSD$mig_stock), bil_mig_UNSD$zeros, bil_mig_UNSD$mig_stock)
bil_mig_UNSD <- select(bil_mig_UNSD, -c("zeros"))

temp <- codes_country
temp <- select(temp, -c(countryname, wpp_region, regioncode)) %>% rename(, "iso3_o" = iso3, "m49code_o" = m49code)
bil_refugees <- merge(x=bil_refugees, y = temp, by =c("iso3_o"), all=TRUE)
temp <- rename(temp, "iso3_d" = iso3_o, "m49code_d" = m49code_o)

bil_refugees <- merge(x=bil_refugees, y = temp, by =c("iso3_d"), all=TRUE) %>% select(, -c(countryname_o, iso3_o, countryname_d, iso3_d))
bil_mig_UNSD <- merge(x=bil_mig_UNSD, y=bil_refugees, by = c("Year", "m49code_o", "m49code_d"), all=TRUE)

bil_mig_UNSD <- bil_mig_UNSD %>% mutate_at(c("refugees_UNHCR"), ~replace(, is.na(.), 0))
bil_mig_UNSD <- bil_mig_UNSD %>% mutate(noref_mig_stock2 = mig_stock-refugees_UNHCR)
bil_mig_UNSD$noref_mig_stock <- ifelse(bil_mig_UNSD$noref_mig_stock2 < 0, 0, bil_mig_UNSD$noref_mig_stock2)
bil_mig_UNSD <- select(bil_mig_UNSD, -c(noref_mig_stock2))

```

Résultat : base de données des stocks de migrants

- Stocks de migrants (réfugiés ou non)
- Stocks bilatéraux de réfugiés

Structure du code VI

Objectif : Flux de migrants

- Différence entre 2 périodes des stocks de migrants
- Deux traitements possibles
 - Pas de migration de retour : les valeurs négatives $\Rightarrow 0$
 - Migration de retour : valeurs négatives $\Rightarrow$ migration entre destination et origine

$$\begin{cases} VAR1_{ij,t} &= stock_{ij,t+1} - stock_{ij,t} \text{ if } stock_{ij,t+1} > stock_{ij,t} \\ VAR1_{ij,t} &= 0 \text{ if } stock_{ij,t+1} < stock_{ij,t} \end{cases}$$

$$\begin{cases} VAR2_{ij,t} &= stock_{ij,t+1} - stock_{ij,t} \text{ if } stock_{ij,t+1} > stock_{ij,t} \& stock_{ji,t+1} > stock_{ji,t} \\ VAR2_{ij,t} &= 0 \text{ if } stock_{ij,t+1} < stock_{ij,t} \& stock_{ji,t+1} > stock_{ji,t} \\ VAR2_{ji,t} &= (stock_{ji,t+1} - stock_{ji,t}) \text{ if } stock_{ji,t+1} > stock_{ji,t} \& stock_{ij,t+1} > stock_{ij,t} \\ VAR2_{ji,t} &= (stock_{ji,t+1} - stock_{ji,t}) + (stock_{ij,t} - stock_{ij,t+1}) \text{ if } stock_{ji,t+1} > stock_{ji,t} \& stock_{ij,t+1} < stock_{ij,t} \end{cases}$$

Structure du code VII

```
# Now we define the flows of migration (with and without refugees) according to two variables : sd_drop_neg and sd_rev_neg

bil_mig_UNSD$id_time <- bil_mig_UNSD %>% group_by(Year) %>% group_indices(Year)#Generate a time index
bil_mig_UNSD$id_o <- bil_mig_UNSD %>% group_by(m49code_o) %>% group_indices(m49code_o)#Generate an origin index
bil_mig_UNSD$id_d <- bil_mig_UNSD %>% group_by(m49code_d) %>% group_indices(m49code_d)#Generate a destination index

library(data.table)
bil_mig_UNSD <- bil_mig_UNSD[order(bil_mig_UNSD$id_o, bil_mig_UNSD$id_d, bil_mig_UNSD$id_time), ]
bil_mig_UNSD$id_bil <- bil_mig_UNSD %>% group_by_at(vars(m49code_o, m49code_d)) %>% group_indices(m49code_o, m49code_d)#Generate a bilateral index

library(dplyr)
#With the refugees negative set to 0
bil_mig_UNSD <- bil_mig_UNSD[order(bil_mig_UNSD$id_o, bil_mig_UNSD$id_d, bil_mig_UNSD$id_time), ]
bil_mig_UNSD <- bil_mig_UNSD %>% group_by(id_bil) %>% mutate(VAR0 = mig_stock - lag(mig_stock, n = 1, order_by = id_bil))
bil_mig_UNSD <- bil_mig_UNSD %>% group_by(id_bil) %>% mutate(sd_drop_neg = lead(VAR0, n = 1, order_by = id_bil))
bil_mig_UNSD$sd_drop_neg <- ifelse(bil_mig_UNSD$sd_drop_neg < 0, 0, bil_mig_UNSD$sd_drop_neg)

# With the refugees : Alternative : reverse migration for negative flows, sd_rev_neg
temp <- bil_mig_UNSD %>% mutate(id_rev_o = id_d) %>% mutate(id_rev_d = id_o)
temp <- select (temp, c(id_rev_d, id_rev_o, VAR0, Year))
temp$REV_VAR0 <- ifelse(temp$VAR0<0, -temp$VAR0, 0)
temp <- select (temp, -c(VAR0))
temp <- rename(temp, "id_d" = id_rev_d, "id_o" = id_rev_o)
bil_mig_UNSD <- merge(x = temp, y = bil_mig_UNSD, by = c("id_d", "id_o", "Year"))

bil_mig_UNSD$TOT_VAR0 <- ifelse(bil_mig_UNSD$VAR0 < 0, bil_mig_UNSD$REV_VAR0, bil_mig_UNSD$VAR0 + bil_mig_UNSD$REV_VAR0)

bil_mig_UNSD <- bil_mig_UNSD[order(bil_mig_UNSD$id_o, bil_mig_UNSD$id_d, bil_mig_UNSD$id_time), ]
bil_mig_UNSD <- bil_mig_UNSD %>% group_by(id_bil.y) %>% mutate(sd_rev_neg = lead(TOT_VAR0, n = 1, order_by = id_bil.y))

bil_mig_UNSD <- select (bil_mig_UNSD, -c(id_bil.x, VAR0, TOT_VAR0, REV_VAR0))
bil_mig_UNSD <- rename(bil_mig_UNSD, "id_bil"=id_bil.y)
```

Résultat : Base de données par pays

Agrégation par région I

Les données économiques

#Version with `gdp_pop_wdi` : `gdp` values by country instead of `gdp per capita`. It will make more sens to sum them by `regioncode`

```
library("dplyr")
country_info <- select (codes_country, -c(wpp_region, countryname, m49code))
region_econ <- rename(country_econ, "iso3" = countrycode)
region_econ <- select(region_econ, -c(countryname))
region_econ <- merge(x = region_econ, y = country_info, by = c("iso3"))
region_econ <- region_econ[!is.na(region_econ$gdp_ppp), ]
region_econ <- region_econ[!is.na(region_econ$pop), ]

library("doBy")
region_econ <- summaryBy(gdp_ppp+pop~Year+regioncode, FUN = c(sum), data = region_econ)
region_econ <- region_econ %>% mutate(gdp_pc = gdp_ppp.sum/pop.sum)

library("purrr")
p5 <- c(0.2, 0.4, 0.6, 0.8)
p5_names <- map_chr(p5, ~paste0(.x*100,"quint"))
p5_funs <- map(p5, ~partial(quantile, probs = .x, na.rm = TRUE)) %>%
  set_names(nm = p5_names)
df.temp <- region_econ %>% group_by(Year) %>% summarize_at(vars(gdp_pc), funs(!p5_funs))
region_econ <- merge(x = region_econ, y = df.temp, by = c("Year"))

region_econ$gdp_low_20 <- ifelse(region_econ$gdp_pc < region_econ$"20quint", 1, 0)
region_econ$gdp_low_40 <- ifelse(region_econ$gdp_pc < region_econ$"40quint", 1, 0)
region_econ$gdp_low_60 <- ifelse(region_econ$gdp_pc < region_econ$"60quint", 1, 0)
region_econ$gdp_low_80 <- ifelse(region_econ$gdp_pc < region_econ$"80quint", 1, 0)

region_econ <- region_econ %>% mutate(RANK1="")
region_econ$RANK1 <- ifelse(region_econ$gdp_pc > region_econ$"80quint", 5,
  ifelse(region_econ$gdp_pc > region_econ$"60quint",
    &region_econ$gdp_pc < region_econ$"80quint", 4,
    ifelse(region_econ$gdp_pc > region_econ$"40quint",
      &region_econ$gdp_pc < region_econ$"60quint", 3,
      ifelse(region_econ$gdp_pc < region_econ$"40quint",
        &region_econ$gdp_pc > region_econ$"20quint", 2,
        ifelse(region_econ$gdp_pc < region_econ$"20quint", 1, region_econ$RANK1))))))
region_econ$RANK1 <- as.numeric(region_econ$RANK1)
```

Agrégation par région II

Les données dyadiques

```
# Generate the bilateral dataset with dyadic values at country level with regioncode

# Step 2.1.1 : Method 1 : dyadic dummies for contiguity, common official language and colonial link between the regions
codes_country_o <- codes_country
colnames(codes_country_o) <- paste(colnames(codes_country), "o", sep = "_")
country_dyadic <- merge(x=dyadic, y=codes_country_o, by = c("iso3_o"), all=TRUE)
codes_country_d <- codes_country
colnames(codes_country_d) <- paste(colnames(codes_country), "d", sep = "_")
country_dyadic <- merge(x=country_dyadic, y=codes_country_d, by = c("iso3_d"), all=TRUE)

library("doBy")
df.temp <- summaryBy(contig + comlang_off + comlang_ethno + colony ~regioncode_d+regioncode_o, FUN = c(sum), data = country_dyadic)
df.temp$contig_region <- ifelse(df.temp$contig.sum == 0, 0, 1)
df.temp$comlang_region <- ifelse(df.temp$comlang_off.sum == 0, 0, 1)
df.temp$comlang2_region <- ifelse(df.temp$comlang_ethno.sum == 0, 0, 1)
df.temp$colony_region <- ifelse(df.temp$colony.sum == 0, 0, 1)

# Step 2.2.1 : Define the distance between a country and a region as the smaller distance
# between that country and the countries from another region
# Define the distance between two regions as the mean of the countries distance to a region
df.temp <- summaryBy(dist+distcap+distw+distwces+distw_comp+distwces_comp
  ~ iso3_o+regioncode_d+regioncode_o,
  FUN = c(min), data = country_dyadic)

df.temp <- summaryBy(dist.min+distcap.min+distw.min+distwces.min+distw_comp.min+distwces_comp.min
  ~ regioncode_d+regioncode_o,
  FUN = c(mean), data = df.temp)

# Step 2.2.2 : Define the distance between a country and a region as the mean distance
# between that country and the countries from another region
region_dist <- summaryBy(dist+distcap+distw+distwces+distw_comp+distwces_comp~iso3_o+regioncode_d+regioncode_o,
  FUN = c(mean), data = country_dyadic)
region_dist <- summaryBy(dist.mean+distcap.mean+distw.mean+distwces.mean+distw_comp.mean+distwces_comp.mean
  ~ regioncode_d+regioncode_o, FUN = c(mean), data = region_dist)

library("dplyr")
region_dist <- merge(x = df.temp, y = region_dist, by = c("regioncode_d", "regioncode_o"))

# We obtain all the dyadic values in one file
library("dplyr")
region_dyadic <- merge(x = region_dist, y = region_dyadic, by = c("regioncode_d", "regioncode_o"))
```

Agrégation par région III

Les données migratoires

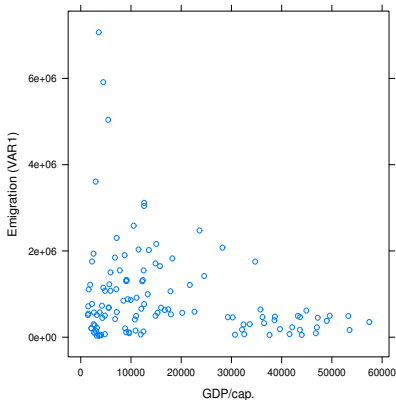
```
region_mig <- summaryBy(sd_drop_neg + sd_rev_neg
  + sd_drop_neg2 + sd_rev_neg2
  + mig_stock + refugees_UNHCR + noref_mig_stock
  ~ regioncode_d+regioncode_o+Year,
  FUN = c(sum), data = complete_data)
```

Pour résumer

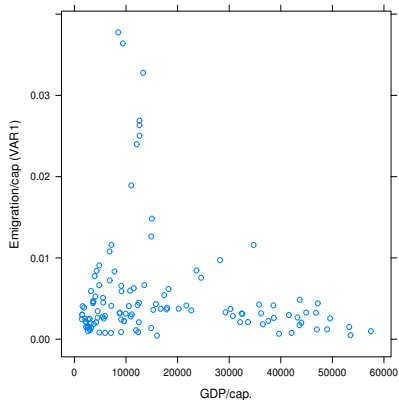
- Acquisition des données par pays
- Agrégation des valeurs par région
- Variables clés :
 - Migration : Variable 1 (sd_drop_neg) et 2 (sd_rev_neg)
 - PIB par habitant - population
 - Variables dyadiques

Faits stylisés V

Figure 14 – GDP per capita and emigration (variable 1)



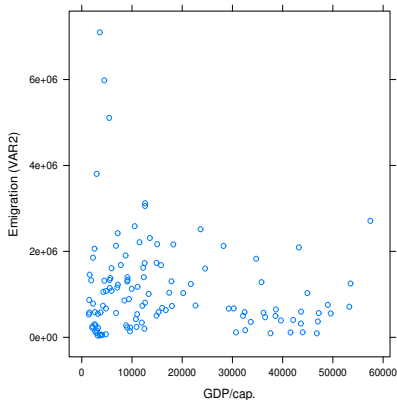
(a) Emigration flows with negative values set to 0
vs. GDP per capita



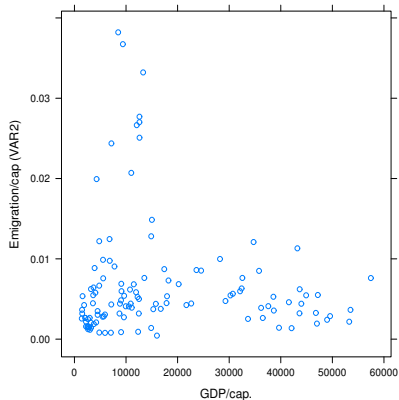
(b) Emigration flows (per capita) with negative values set to 0 vs. GDP per capita

Faits stylisés VI

Figure 16 – GDP per capita and emigration (variable 2)



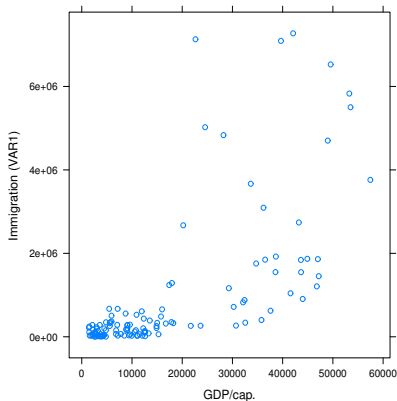
(a) Emigration flows with reverse migration vs. GDP per capita



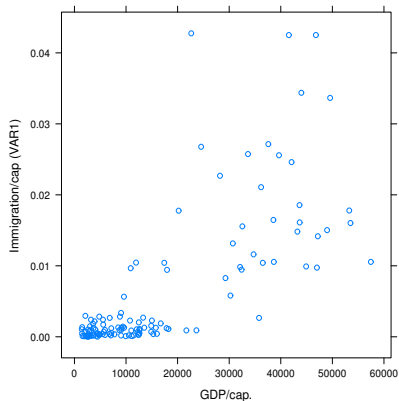
(b) Emigration flows per capita with reverse migration vs. GDP per capita

Faits stylisés VII

Figure 18 – GDP per capita and immigration (variable 1)



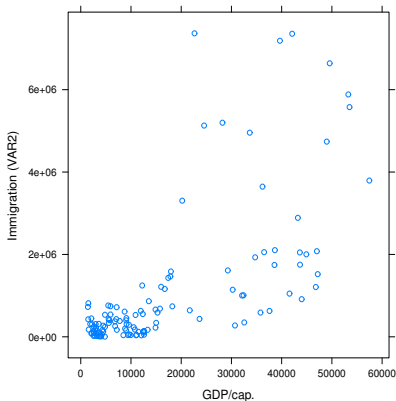
(a) Immigration flows with negative values set to 0 vs. GDP per capita



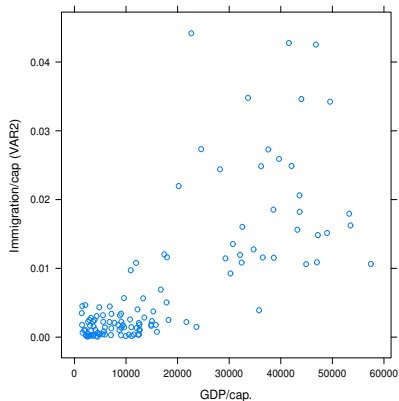
(b) Immigration flows (per capita) with negative values set to 0 vs. GDP per capita

Faits stylisés VIII

Figure 20 – GDP per capita and immigration (variable 2)



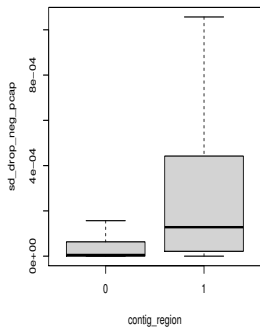
(a) Immigration flows with reverse migration vs. GDP per capita



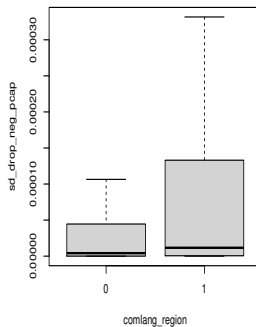
(b) Immigration flows per capita with reverse migration vs. GDP per capita

Faits stylisés IX

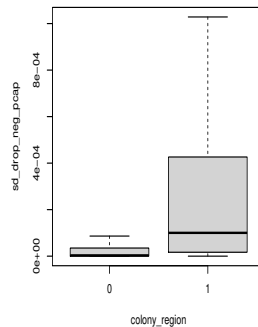
Figure 22 – Dyadic variables (variable 1)



(a) Contiguïté



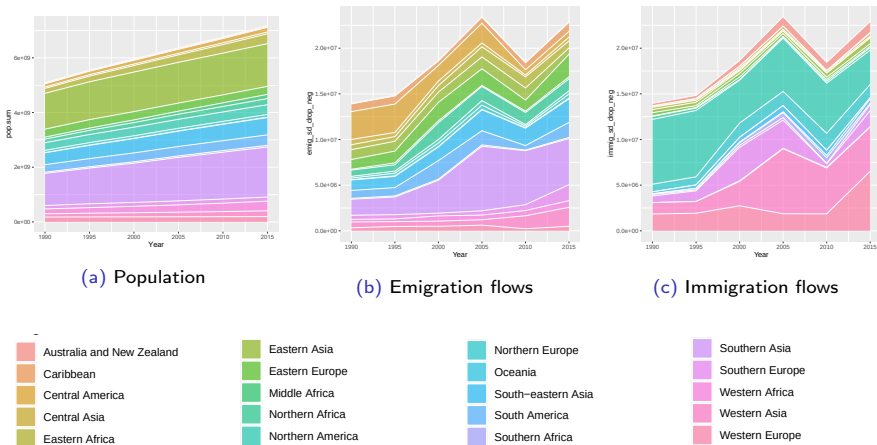
(b) Langue officielle



(c) Lien colonial

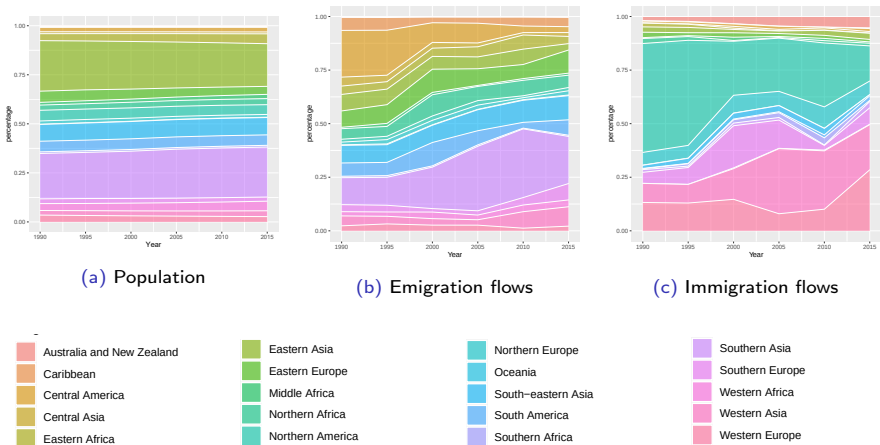
Faits stylisés X

Figure 24 – Migration flows



Faits stylisés XI

Figure 26 – Migration flows in terms of global share



1

Motivations

- Le projet Navigate - Modèle initial
- Littérature

2

Base de données

- Construction de la base de données
- Faits stylisés

3

Analyse économétrique

- Méthodologie
- Résultats
- Tests de robustesse

4

Conclusion

Méthodologie

Estimation Scaled-OLS avec effets fixes pour le temps

- Construction des variables en log
- Division de la base de données en sous-échantillons
- Équation

$$\begin{aligned} \ln(1 + emig_{od}) = & Constant + \beta_1 \ln(pop_o) + \beta_2 \ln(pop_d) \\ & + \beta_3 \ln(gdppc_o) + \beta_4 \ln(gdppc_d) \\ & + \beta_5 distw_{od}^a + \beta_6 contiguity_{od} + \beta_7 offlang_{od} + \beta_8 colony_{od} + \varepsilon_t \end{aligned}$$

- Tests de robustesse : Effets fixes origine, SIDS, Réfugiés

```

SOLS_FE_pop_VAR1_o2111_21 <- lm(log_var10 ~ log_pop_o + log_pop_d + log_gdppc_o + log_gdppc_d + log_distw_comp_min_mean
+ contig_region + comlang_region + colony_region + factor(id_time),
data=UNSD_region_mig_o21)

SOLS_FE_pop_VAR1_o2111_22 <- lm(log_var10 ~ log_pop_o + log_pop_d + log_gdppc_o + log_gdppc_d + log_distw_comp_min_mean
+ contig_region + comlang_region + colony_region + factor(id_time),
data=UNSD_region_mig_o22)

AGG_SOLS_FE_pop_VAR1_o2111_21 <- UNSD_region_mig_o21 %>% mutate(., tidy(SOLS_FE_pop_VAR1_o2111_21)
%>% dplyr::select(term, estimate) %>% spread(term, estimate)
%>% rename_all(toupper))
AGG_SOLS_FE_pop_VAR1_o2111_21 <- AGG_SOLS_FE_pop_VAR1_o2111_21 %>% mutate(., tidy(SOLS_FE_pop_VAR1_o2111_21)
%>% dplyr::select(term, p.value)
%>% spread(term, p.value)
%>% rename_all(~ c(paste0("PVAL", 1:14))))

AGG_SOLS_FE_pop_VAR1_o2111_22 <- UNSD_region_mig_o22 %>% mutate(., tidy(SOLS_FE_pop_VAR1_o2111_22)
%>% dplyr::select(term, estimate)
%>% spread(term, estimate)
%>% rename_all(toupper))
AGG_SOLS_FE_pop_VAR1_o2111_22 <- AGG_SOLS_FE_pop_VAR1_o2111_22 %>% mutate(., tidy(SOLS_FE_pop_VAR1_o2111_22)
%>% dplyr::select(term, p.value)
%>% spread(term, p.value)
%>% rename_all(~ c(paste0("PVAL", 1:14))))

AGG_SOLS_FE_pop_VAR1_o2111 <- rbindlist(list(AGG_SOLS_FE_pop_VAR1_o2111_21, AGG_SOLS_FE_pop_VAR1_o2111_22), fill=TRUE)
rm(AGG_SOLS_FE_pop_VAR1_o2111_21, AGG_SOLS_FE_pop_VAR1_o2111_22)

```

```

SIGN_COEF <- paste0("SIGN_", names(AGG_SOLS_FE_pop_VAR1_o2111[, c(113:126)]))
COEF <- paste0(names(AGG_SOLS_FE_pop_VAR1_o2111[, c(113:126)]))
PVAL <- paste0(names(AGG_SOLS_FE_pop_VAR1_o2111[, c(127:140)]))

library("data.table")
for(i in 1:14){
  a=get("i")
  b=SIGN_COEF[[a]]
  c=PVAL[[a]]
  d=COEF[[a]]
  AGG_SOLS_FE_pop_VAR1_o2111[[b]]<- ifelse(AGG_SOLS_FE_pop_VAR1_o2111[[c]]<0.1, AGG_SOLS_FE_pop_VAR1_o2111[[d]], 0)
}

AGG_SOLS_FE_pop_VAR1_o2111 <- AGG_SOLS_FE_pop_VAR1_o2111 %>% ungroup()
AGG_SOLS_FE_pop_VAR1_o2111 <- AGG_SOLS_FE_pop_VAR1_o2111 %>% mutate(., FITTED = exp(AGG_SOLS_FE_pop_VAR1_o2111$'SIGN_(INTERCEPT)')*
exp(id1995*AGG_SOLS_FE_pop_VAR1_o2111$'SIGN_FACTOR(ID_TIME)2')*
exp(id2000*AGG_SOLS_FE_pop_VAR1_o2111$'SIGN_FACTOR(ID_TIME)3')*
exp(id2005*AGG_SOLS_FE_pop_VAR1_o2111$'SIGN_FACTOR(ID_TIME)4')*
exp(id2010*AGG_SOLS_FE_pop_VAR1_o2111$'SIGN_FACTOR(ID_TIME)5')*
exp(id2015*AGG_SOLS_FE_pop_VAR1_o2111$'SIGN_FACTOR(ID_TIME)6')*
pop.sum_o^AGG_SOLS_FE_pop_VAR1_o2111$'SIGN_LOG_POP_O'*
pop.sum_d^AGG_SOLS_FE_pop_VAR1_o2111$'SIGN_LOG_POP_D'*
gdp.pc_o^AGG_SOLS_FE_pop_VAR1_o2111$'SIGN_LOG_GDPPC_O'*
gdp.pc_d^AGG_SOLS_FE_pop_VAR1_o2111$'SIGN_LOG_GDPPC_D'*
distw.comp.min.mean^AGG_SOLS_FE_pop_VAR1_o2111$'SIGN_LOG_DISTW_COM'*
exp(contig_region^AGG_SOLS_FE_pop_VAR1_o2111$'SIGN_CONTIG_REGION')*
exp(comlang_region^AGG_SOLS_FE_pop_VAR1_o2111$'SIGN_COMLANG_REGION')*
exp(colony_region^AGG_SOLS_FE_pop_VAR1_o2111$'SIGN_COLONY_REGION')

```

Résultats de la régression I

Table 1 – Population and negative values set to 0

	No groups	Low-inc	High-inc
(Intercept)	−34.5353*** (1.6577)	−31.2724*** (2.1432)	−41.5367*** (3.4204)
Orig. Pop.	0.9249*** (0.0396)	0.9478*** (0.0429)	0.7089*** (0.0972)
Dest. Pop.	0.3850*** (0.0397)	0.2915*** (0.0516)	0.4739*** (0.0556)
Orig. GDP per capita	0.5177*** (0.0526)	0.7173*** (0.1014)	1.2451*** (0.1748)
Dest. GDP per capita	2.2873*** (0.0528)	2.5608*** (0.0752)	2.1032*** (0.0742)
Distances	−1.2524*** (0.0942)	−1.9498*** (0.1318)	−0.8064*** (0.1236)
Contiguity	1.9133*** (0.1843)	2.5477*** (0.2406)	1.1106*** (0.2786)
Common off. language	1.7278*** (0.1167)	1.8764*** (0.1540)	1.1313*** (0.1659)
Colonial Link	0.8714*** (0.1343)	0.9918*** (0.2067)	0.8947*** (0.1686)
Time Fixed Effects	YES	YES	YES
N	2280	1368	912
R ²	0.6532	0.6984	0.6469
adj. R ²	0.6513	0.6955	0.6418
Resid. sd	2.3281	2.3195	2.0811

Standard errors in parentheses

† significant at $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$

Résultats de la régression II

Table 2 – Population and negative values as reverse migration

	No groups	Low-inc	High-inc
(Intercept)	−32.0617*** (1.5204)	−31.2794*** (1.9272)	−37.0329*** (3.1774)
Orig. Pop.	0.8003*** (0.0363)	0.7918*** (0.0386)	0.7242*** (0.0903)
Dest. Pop.	0.6151*** (0.0364)	0.5910*** (0.0464)	0.6084*** (0.0517)
Orig. GDP per capita	0.6375*** (0.0482)	0.9081*** (0.0912)	1.2098*** (0.1624)
Dest. GDP per capita	1.8900*** (0.0484)	2.2039*** (0.0676)	1.6394*** (0.0690)
Distances	−1.3436*** (0.0864)	−1.9866*** (0.1185)	−0.9312*** (0.1148)
Contiguity	1.7918*** (0.1690)	2.4320*** (0.2164)	0.9459*** (0.2588)
Common off. language	1.5170*** (0.1070)	1.9554*** (0.1385)	0.6230*** (0.1541)
Colonial Link	0.9087*** (0.1232)	1.0245*** (0.1858)	0.9296*** (0.1566)
<i>N</i>	2280	1368	912
<i>R</i> ²	0.6558	0.7118	0.6288
adj. <i>R</i> ²	0.6538	0.7091	0.6234
Resid. sd	2.1354	2.0857	1.9332

Standard errors in parentheses

† significant at $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$

Résultats de la régression III

- Tous les coefficients des variables indépendantes sont significatifs, et les régressions semblent être assez significatives
- Différences importantes entre les deux groupes dans les valeurs des coefficients, mais pas dans les signes $\Rightarrow$ Relation non linéaire
- Les distances reflètent le coût de la migration $\Rightarrow$ Problème pour les résidents des régions pauvres
- Effet du revenu du pays d'origine plus fort pour les régions riches.
- Effet du revenu du pays de destination plus fort pour les régions moins riches
- Effets de la taille de la population différents selon le sous-groupe

Résultats de la régression IV

Table 3 – Population and negative values set to 0

	No groups	low to low inc.	low to high inc.	high to high inc.	high to low inc.
(Intercept)	-34.5353*** (1.6577)	-24.5939*** (3.1768)	-47.2419*** (2.9616)	-34.9922*** (5.3212)	-42.4357*** (4.5046)
Orig. Pop.	0.9249*** (0.0396)	0.9406*** (0.0622)	0.9252*** (0.0414)	0.6165*** (0.0999)	0.7472*** (0.1318)
Dest. Pop.	0.3850*** (0.0397)	0.2911*** (0.0623)	0.2192** (0.0842)	0.1735† (0.1004)	0.5262*** (0.0654)
Orig. GDP per capita	0.5177*** (0.0526)	0.6965*** (0.1487)	0.5826*** (0.0975)	0.4153** (0.1603)	1.8443*** (0.2614)
Dest. GDP per capita	2.2873*** (0.0528)	1.4266*** (0.1509)	4.5106*** (0.1658)	2.7907*** (0.1628)	1.8042*** (0.1549)
Distances	-1.2524*** (0.0942)	-1.6346*** (0.2144)	-1.9387*** (0.1151)	-0.4184** (0.1415)	-1.3769*** (0.1706)
Contiguity	1.9133*** (0.1843)	3.0142*** (0.3444)	2.4030*** (0.3041)	1.4008*** (0.2627)	1.4490** (0.4878)
Common off. language	1.7278*** (0.1167)	2.2681*** (0.2290)	0.8885*** (0.1486)	0.3820* (0.1620)	1.5040*** (0.2338)
Colonial Link	0.8714*** (0.1343)	-1.4756† (0.7614)	0.4164** (0.1494)	0.3001† (0.1675)	1.2187*** (0.2370)
Time Fixed Effects	YES	YES	YES	YES	YES
N	2280	792	576	336	576
R ²	0.6532	0.5675	0.8005	0.7069	0.5064
adj. R ²	0.6513	0.5603	0.7959	0.6950	0.4950
Resid. sd	2.3281	2.5564	1.4418	1.1662	2.2714

Standard errors in parentheses

† significant at $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$

Résultats de la régression V

Table 4 – Population and negative values as reverse migration

	No groups	low to low inc.	low to high inc.	high to high inc.	high to low inc.
(Intercept)	−32.0617*** (1.5204)	−25.1691*** (2.7170)	−46.9162*** (3.0388)	−45.6612*** (5.5186)	−33.3415*** (4.0441)
Orig. Pop.	0.8003*** (0.0363)	0.7067*** (0.0532)	0.8824*** (0.0425)	0.6730*** (0.1037)	0.7817*** (0.1183)
Dest. Pop.	0.6151*** (0.0364)	0.6277*** (0.0533)	0.3003*** (0.0864)	0.4625*** (0.1041)	0.6549*** (0.0588)
Orig. GDP per capita	0.6375*** (0.0482)	0.8984*** (0.1271)	0.7889*** (0.1001)	0.9450*** (0.1662)	1.5569*** (0.2347)
Dest. GDP per capita	1.8900*** (0.0484)	1.3083*** (0.1291)	4.2658*** (0.1701)	2.4767*** (0.1689)	1.2434*** (0.1391)
Distances	−1.3436*** (0.0864)	−1.7607*** (0.1834)	−1.9308*** (0.1181)	−0.1161 (0.1467)	−1.6386*** (0.1532)
Contiguity	1.7918*** (0.1690)	2.8800*** (0.2946)	2.2628*** (0.3120)	1.8371*** (0.2724)	1.4338** (0.4380)
Common off. language	1.5170*** (0.1070)	2.4139*** (0.1959)	0.7829*** (0.1524)	0.1076 (0.1680)	0.7916*** (0.2099)
Colonial Link	0.9087*** (0.1232)	0.9002 (0.6512)	0.4249** (0.1533)	0.5295** (0.1737)	1.3121*** (0.2127)
Time Fixed Effects	YES	YES	YES	YES	YES
N	2280	792	576	336	576
R ²	0.6558	0.6422	0.7794	0.6631	0.5433
adj. R ²	0.6538	0.6362	0.7743	0.6495	0.5327
Resid. sd	2.1354	2.1864	1.4794	1.2095	2.0391

Standard errors in parentheses

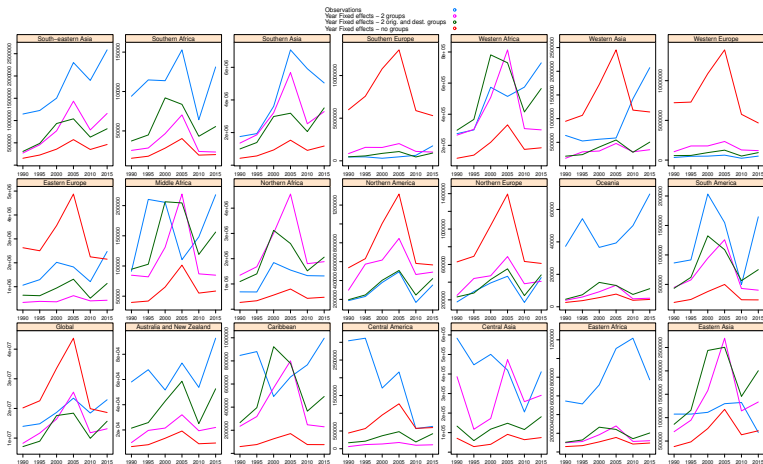
† significant at $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$

Résultats de la régression VI

- Toutes les régressions sont assez significatives, à l'exception de la régression de la migration entre les régions à revenu élevé et à faible revenu
- Les résultats les plus significatifs : migration entre les régions à faible revenu et à revenu élevé
- La plupart des coefficients des variables indépendantes sont significatifs, même dans la quatrième régression
- Le PIB par habitant des pays d'origine a un impact plus fort sur la migration vers les pays à faible revenu, que les régions d'origine soient riches ou non
- Le gain de la migration a un effet très fort pour la migration des régions à faible revenu vers les régions à revenu élevé
- Les variables dyadiques comptent peu pour la migration entre régions riches, et beaucoup pour la migration venant des pays pauvres
- Impact négatif du lien colonial pour la migration entre régions pauvres
- Les résultats restent les mêmes pour la variable 2, cependant ils sont moins significatifs

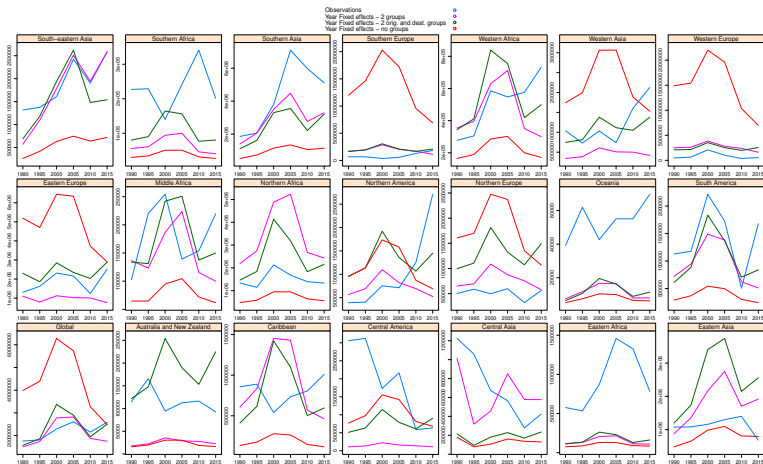
Représentations graphiques - prédictions du modèle I

Figure 28 – Emigration flows by region (negative values set to 0 - variable 1)



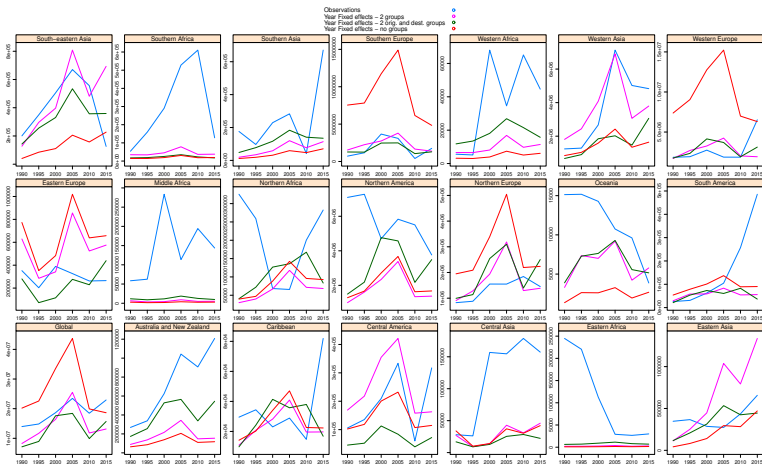
Représentations graphiques - prédictions du modèle II

Figure 29 – Emigration flows by region (negative values as reverse migration - variable 2)



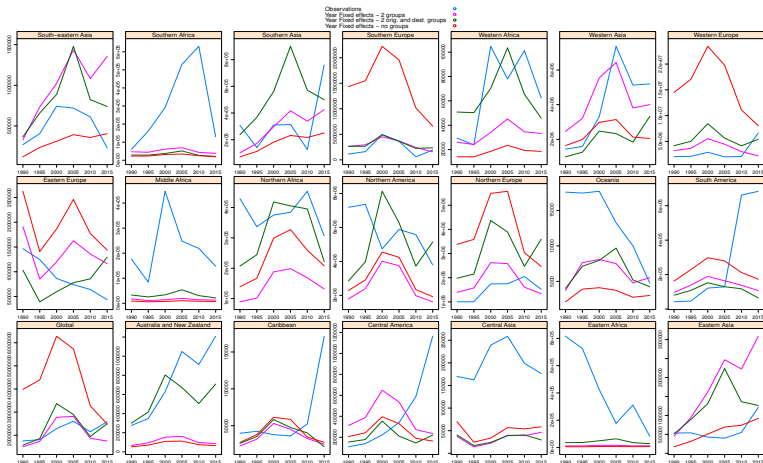
Représentations graphiques - prédictions du modèle III

Figure 30 – Immigration flows by region (negative values set to 0 - variable 1)



Représentations graphiques - prédictions du modèle IV

Figure 31 – Immigration flows by region (negative values as reverse migration - variable 2)



Tests de robustesse I

Table 5 – Population and negative values set to 0

	No groups	Low-inc	High-inc
(Intercept)	−33.8363*** (1.7633)	−31.7697*** (2.1715)	−42.8749*** (3.3706)
Orig. Pop.	0.8422*** (0.0511)	0.9507*** (0.0615)	0.6443*** (0.0962)
Dest. Pop.	0.3555*** (0.0390)	0.2475*** (0.0509)	0.4646*** (0.0547)
Orig. GDP per capita	0.7107*** (0.0573)	0.8550*** (0.1022)	1.5899*** (0.1823)
Dest. GDP per capita	2.0271*** (0.0581)	2.3126*** (0.0813)	1.8793*** (0.0829)
Distances	−1.0402*** (0.0950)	−1.7046*** (0.1363)	−0.6686*** (0.1239)
Contiguity	1.7442*** (0.1814)	2.3446*** (0.2370)	1.0569*** (0.2740)
Common off. language	1.6126*** (0.1155)	1.6790*** (0.1554)	1.0353*** (0.1640)
Colonial Link	0.8936*** (0.1318)	0.8636*** (0.2037)	0.9298*** (0.1659)
SIDS	0.1105 (0.2150)	0.6064* (0.2507)	
Ref. Sender	1.2952*** (0.1317)	1.2899*** (0.1721)	1.0945*** (0.1927)
Time Fixed Effects	YES	YES	YES
N	2280	1368	912
R ²	0.6675	0.7114	0.6592
adj. R ²	0.6653	0.7082	0.6539
Resid. sd	2.2809	2.2707	2.0458

Standard errors in parentheses

† significant at $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$

Tests de robustesse II

Table 6 – Population and negative values as reverse migration

	No groups	Low-inc	High-inc
(Intercept)	−32.2022*** (1.6145)	−31.7153*** (1.9524)	−38.2579*** (3.1328)
Orig. Pop.	0.7599*** (0.0468)	0.7934*** (0.0553)	0.6651*** (0.0894)
Dest. Pop.	0.5863*** (0.0357)	0.5513*** (0.0457)	0.5999*** (0.0508)
Orig. GDP per capita	0.8380*** (0.0524)	1.0325*** (0.0919)	1.5253*** (0.1694)
Dest. GDP per capita	1.6449*** (0.0532)	1.9800*** (0.0731)	1.4345*** (0.0771)
Distances	−1.1513*** (0.0870)	−1.7649*** (0.1225)	−0.8051*** (0.1151)
Contiguity	1.6301*** (0.1661)	2.2489*** (0.2131)	0.8967*** (0.2547)
Common off. language	1.3946*** (0.1057)	1.7781*** (0.1398)	0.5352*** (0.1524)
Colonial Link	0.9205*** (0.1207)	0.9094*** (0.1832)	0.9617*** (0.1542)
SIDS_o	0.3564† (0.1969)	0.5412* (0.2254)	
Ref. Sender	1.2240*** (0.1206)	1.1633*** (0.1548)	1.0020*** (0.1791)
Time Fixed Effects	YES	YES	YES
N	2280	1368	912
R ²	0.6710	0.7243	0.6413
adj. R ²	0.6688	0.7212	0.6357
Resid. sd	2.0886	2.0417	1.9014

Standard errors in parentheses

† significant at $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$

Tests de robustesse III

Table 7 – Population and negative values set to 0

	No groups	low to low inc.	low to high inc.	high to high inc.	high to low inc.
(Intercept)	-33.8363*** (1.7633)	-26.7250*** (3.2174)	-46.8939*** (3.0114)	-34.6914*** (5.3416)	-44.1913*** (4.4587)
Orig. Pop.	0.8422*** (0.0511)	1.0174*** (0.0913)	0.9573*** (0.0593)	0.6047*** (0.1014)	0.6949*** (0.1305)
Dest. Pop.	0.3555*** (0.0390)	0.2556*** (0.0621)	0.1993* (0.0830)	0.1768† (0.1006)	0.5112*** (0.0646)
Orig. GDP per capita	0.7107*** (0.0573)	0.8211*** (0.1531)	0.6132*** (0.0978)	0.4574** (0.1707)	2.0965*** (0.2645)
Dest. GDP per capita	2.0271*** (0.0581)	1.2935*** (0.1532)	4.3088*** (0.1743)	2.7316*** (0.1825)	1.6212*** (0.1587)
Distances	-1.0402*** (0.0950)	-1.4803*** (0.2258)	-1.8308*** (0.1191)	-0.4180** (0.1416)	-1.1448*** (0.1769)
Contiguity	1.7442*** (0.1814)	2.8284*** (0.3426)	2.4211*** (0.2994)	1.3846*** (0.2639)	1.0963* (0.4880)
Common off. language	1.6126*** (0.1155)	2.0008*** (0.2355)	0.8196*** (0.1495)	0.3889* (0.1624)	1.3954*** (0.2319)
Colonial Link	0.8936*** (0.1318)	-1.2673† (0.7687)	0.3102* (0.1492)	0.2891† (0.1683)	1.3258*** (0.2349)
SIDS_o	0.1105 (0.2150)	0.8916* (0.3736)	0.5367* (0.2393)		
Ref. Sender	1.2952*** (0.1317)	0.9830*** (0.2476)	0.7515*** (0.1874)	0.1270 (0.1767)	1.2533*** (0.2979)
N	2280	792	576	336	576
R ²	0.6675	0.5791	0.8075	0.7073	0.5215
adj. R ²	0.6653	0.5710	0.8023	0.6946	0.5096
Resid. sd	2.2809	2.5252	1.4187	1.1671	2.2383

Standard errors in parentheses

† significant at $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$

Tests de robustesse IV

Table 8 – Population and negative values as reverse migration

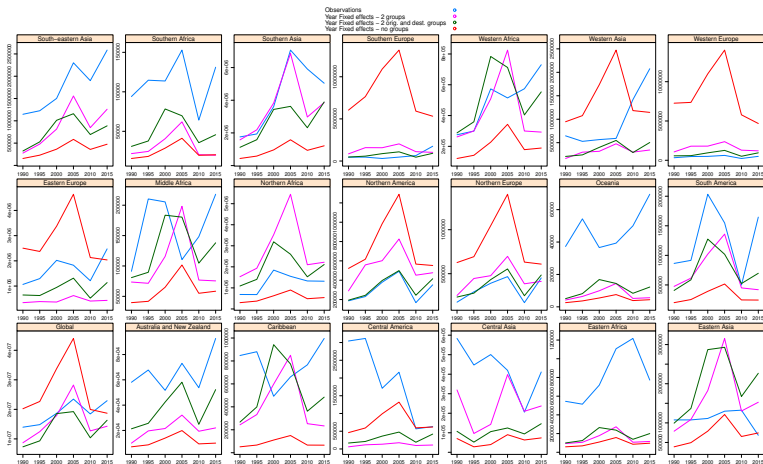
	No groups	low to low inc.	low to high inc.	high to high inc.	high to low inc.
(Intercept)	−32.2022*** (1.6145)	−27.0082*** (2.7419)	−46.8068*** (3.1101)	−45.8739*** (5.5422)	−34.9268*** (4.0021)
Orig. Pop.	0.7599*** (0.0468)	0.7559*** (0.0778)	0.9185*** (0.0613)	0.6814*** (0.1052)	0.7345*** (0.1171)
Dest. Pop.	0.5863*** (0.0357)	0.5928*** (0.0529)	0.2833** (0.0858)	0.4602*** (0.1044)	0.6414*** (0.0580)
Orig. GDP per capita	0.8380*** (0.0524)	1.0303*** (0.1305)	0.8100*** (0.1010)	0.9153*** (0.1771)	1.7847*** (0.2374)
Dest. GDP per capita	1.6449*** (0.0532)	1.1735*** (0.1305)	4.1104*** (0.1800)	2.5184*** (0.1893)	1.0781*** (0.1425)
Distances	−1.1513*** (0.0870)	−1.5868*** (0.1925)	−1.8489*** (0.1230)	−0.1165 (0.1469)	−1.4290*** (0.1588)
Contiguity	1.6301*** (0.1661)	2.7039*** (0.2920)	2.2801*** (0.3092)	1.8486*** (0.2738)	1.1153* (0.4380)
Common off. language	1.3946*** (0.1057)	2.1695*** (0.2007)	0.7201*** (0.1544)	0.1028 (0.1685)	0.6936*** (0.2081)
Colonial Link	0.9205*** (0.1207)	1.1560† (0.6551)	0.3355* (0.1541)	0.5372** (0.1746)	1.4088*** (0.2108)
SIDS_o	0.3564† (0.1969)	0.7347* (0.3184)	0.4881* (0.2472)		
Ref. Sender	1.2240*** (0.1206)	0.9875*** (0.2110)	0.5955** (0.1935)	−0.0898 (0.1833)	1.1317*** (0.2674)
Time Fixed Effects	YES	YES	YES	YES	YES
N	2280	792	576	336	576
R ²	0.6710	0.6543	0.7844	0.6633	0.5574
adj. R ²	0.6688	0.6476	0.7786	0.6487	0.5464
Resid. sd	2.0886	2.1520	1.4652	1.2109	2.0091

Standard errors in parentheses

† significant at $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$

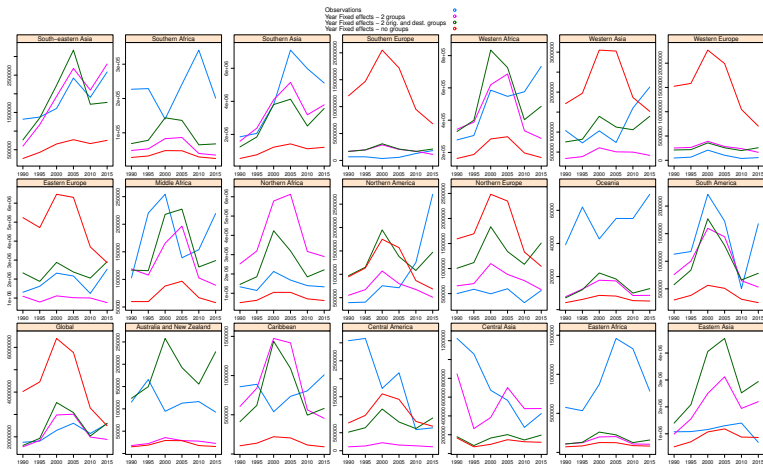
Tests de robustesse V

Figure 32 – Emigration flows by region (negative values set to 0 - variable 1)



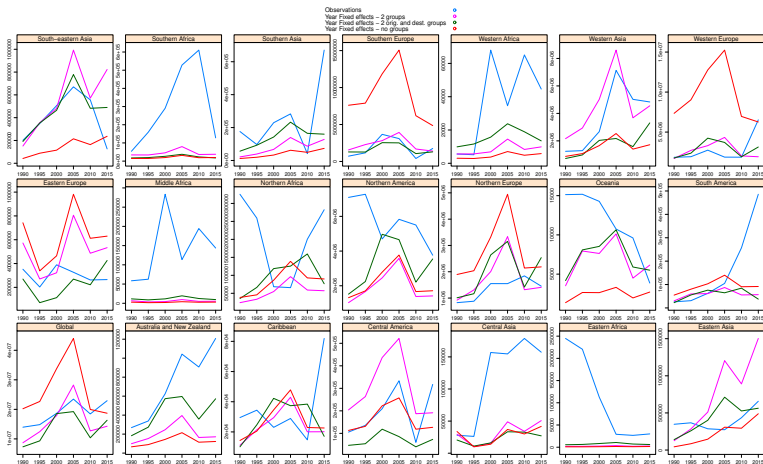
Tests de robustesse VI

Figure 33 – Emigration flows by region (negative values as reverse migration - variable 2)



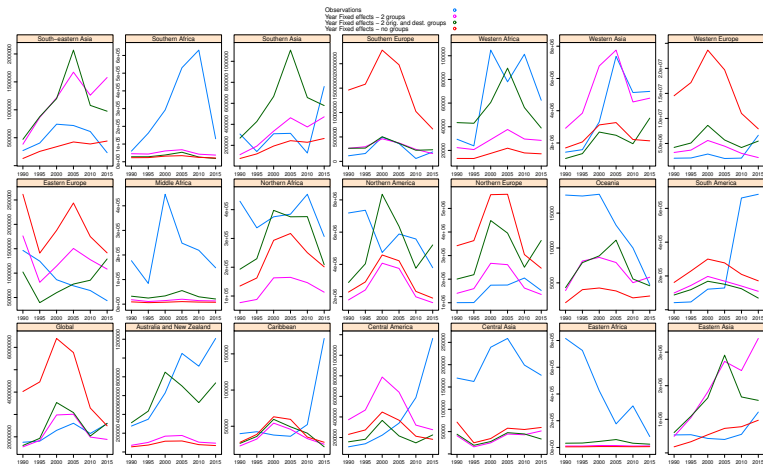
Tests de robustesse VII

Figure 34 – Immigration flows by region (negative values set to 0 - variable 1)



Tests de robustesse VIII

Figure 35 – Immigration flows by region (negative values as reverse migration - variable 2)



- 1 Motivations
 - Le projet Navigate - Modèle initial
 - Littérature
- 2 Base de données
 - Construction de la base de données
 - Faits stylisés
- 3 Analyse économétrique
 - Méthodologie
 - Résultats
 - Tests de robustesse
- 4 Conclusion

Conclusion I

Ce qui marche

- Effets non linéaires : revenus, coûts
- Caractéristiques régionales : proximité culturelle ou historique
- Prédications plutôt justes

Ce qu'il reste à faire

- Modèles économétriques : pseudo-maximum likelihood (PPML), prise en compte de la résistance multilatérale, des variables omises
- Transformation des flux par tranche d'âge et sexe
 - Flux bilatéraux observés (par pays et par régions) et prédits par régions
 - Poids des groupes de population par âge et sexe dans le stock de migrants reçus par pays
 - Stock de migrants dans le pays de destination par âge et sexe

Conclusion II

Méthode 1 : transformation a priori

- En supposant les poids similaires pour toutes les origines, transformer toutes les données bilatérales par pays : sexe et âge
- Agrégation par région des stocks par sexe et âge
- Calcul des flux avec la méthode développée
- Estimation par groupe (qualité des résultats ??)

Méthode 2 : transformation a posteriori

- Estimation et prédiction des flux bilatéraux par région
- Définir le poids des groupes (âge et sexe) à l'échelle régionale : pondérer les pays dans les stocks régionaux et faire une moyenne sur toute la période
- Diviser les flux prédits par groupe en supposant des poids similaires pour toutes les régions d'origine

Merci de votre attention

5 References

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